

Mid-term test

Question 1

(a) Electric field line diverges or the divergence of electric field is not zero but equal to the volume charge density over the permittivity of the medium. 2 marks

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon}$$

~~Taking sigma~~ Taking volume integral on both sides,

$$\int_{Vol} \vec{\nabla} \cdot \vec{E} \, dV = \int_{Vol} \frac{\rho}{\epsilon} \, dV$$

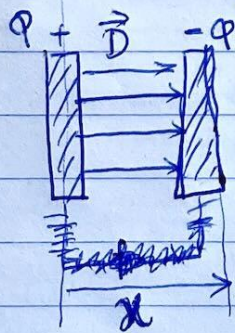
In the HS divergence theorem can be applied,

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q_{enclosed}}{\epsilon}, \quad Q_{enclosed} = \int_{Vol} \rho \, dV$$

This equation is the representation of Gauss's law. 3 marks

(b)

(i) If we assume the plates are large compared to separation distance, the fringing of the flux at the edges of the plates can be neglected. This allows us to consider electric field $\vec{E}$ and density $\vec{D}$ to be perpendicular to the plate. The



inner plate surface can be considered as the Gaussian surface to calculate $\vec{E}$ and $\vec{D}$.

(ii) $V = 25 \text{ kV}$ $d = 1 \text{ cm}$ $S = 250 \text{ cm}^2$

$$\oint \vec{E} \cdot d\vec{l} = E \cdot d \quad \therefore \vec{E} = \frac{V}{d} \hat{x}$$

$$(ii) \quad Q = +5 \mu C \quad d = 1 \text{ mm} \quad S = 250 \text{ cm}^2$$

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

$$E S = \frac{Q_{enc}}{\epsilon_0}$$

$$E = \frac{Q_{enc}}{\epsilon_0 S}$$

$$D = \epsilon_0 E = \frac{Q_{enc}}{S} = \frac{5 \times 10^{-6}}{250 \times 10^{-4}}$$

$$\vec{D} = |D| \vec{a}_x$$

(iii) $\vec{D} = \frac{Q}{S} \vec{a}_x$ which is not function of separation distance d . Hence, the value does not change.

Question 2

$$(a) \quad \vec{E} = \frac{f_L}{2\pi\epsilon_0 r} \vec{a}_r, \quad V_{12} = -\int_1^2 \vec{E} \cdot d\vec{l}$$

$$V_{12} = -\int_{r_1}^{r_2} \frac{f_L}{2\pi\epsilon_0 r} dr$$

$$= -\left[\frac{f_L}{2\pi\epsilon_0} \ln(r) \right]_{r_1}^{r_2}$$

$$= \frac{f_L}{2\pi\epsilon_0} [-\ln r_2 + \ln(r_1)]$$

$$= \frac{f_L}{2\pi\epsilon_0} \ln\left(\frac{r_1}{r_2}\right)$$

$$(ii) \quad r_2 = \frac{1.2}{2} \text{ cm}, \quad r_1 = h = 7 \text{ m}$$

$$V = \frac{q l}{2\pi \epsilon_0} \ln\left(\frac{0.6 \times 10^{-2}}{7}\right)$$

$$= \frac{100 \times 10^{-9}}{2\pi \times 8.854 \times 10^{-12}} \ln\left(\frac{0.6 \times 10^{-2}}{7}\right)$$

$$= -12.69 \text{ kV}$$

(iii) Capacitance / meter between the line and the ground can be calculated using method of image -

$$C = \frac{\pi \epsilon_0}{\ln \sqrt{\frac{d}{2a} + \sqrt{\left(\frac{d}{2a}\right)^2 - 1}}} \text{ F/m}$$

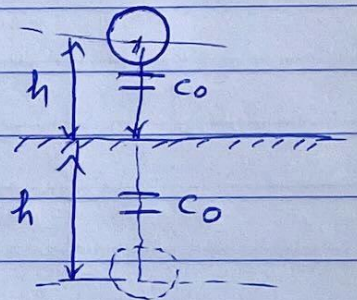
$$d = 2h = 14 \text{ m}$$

$$a = 0.6 \text{ cm}$$

$$1/C = 1/C_0 + 1/C_0$$

$$C = \frac{C_0}{2}$$

$$\therefore C_0 = 2C = \frac{2\pi \epsilon_0}{\ln \sqrt{\frac{14}{1.2 \times 10^{-2}} + \sqrt{\left(\frac{14}{1.2 \times 10^{-2}}\right)^2 - 1}}} \text{ F/m}$$



To calculate capacitance between the conductor and the ground, we cannot ignore the effect of electric field of the ground other on the electric field of the conductor.

Method of image allows us to change the ground plane to an equip. surface and the image of the conductor.

Thus, the problem changes to finding capacitance between two ~~parallel~~ oppositely charged parallel conductors.

The capacitance between the conductor and the ground is half the value of the total capacitance between the conductor and its image as shown in the figure.

(b) For the ~~3~~ 3 conductors and the ground, the capacitance matrix is a 3×3 matrix.

$$C = \begin{bmatrix} (10+5+5) & -5 & -5 \\ -5 & 20 & -5 \\ -5 & -5 & 20 \end{bmatrix} \text{ pF/m}$$

Question 3

(a) The maximum electric field that an insulating material can withstand without a breakdown. When breakdown occurs, bound charges of insulating material detach from the atom and free availability of free charges makes the insulating material behave like a conductor.

(b) $a = 2 \text{ mm}$ outer radius, $b = (2+4) = 6 \text{ mm}$

$$\epsilon = 3\epsilon_0 \text{ F/m} \quad E_{\text{max}} = 25 \text{ MV/m}$$

(c)

The electric field at $r = a$ is the maximum. If this value is lower than 25 MV/m , breakdown can be prevented.

$$E_{\text{max}} = \frac{q_l}{2\pi\epsilon a} \leq 25 \text{ MV/m}$$

$$q_l = 25 \times 10^6 (2\pi \times 3\epsilon_0 \times 2 \times 10^{-3}) \text{ C/m}$$

The maximum voltage V_0 can be found as $V_{ab} = - \int_b^a \vec{E} \cdot d\vec{l}$

$$V = \frac{\rho l}{2\pi\epsilon} \ln\left(\frac{b}{a}\right)$$

$$= \frac{\rho}{2\pi \times 3\epsilon_0} \ln\left(\frac{6}{2}\right) \text{ Volt} =$$

(ii) Capacitance $C = \frac{2\pi\epsilon}{\ln(b/a)} \text{ F/m}$

$$= \frac{2\pi \times 3\epsilon_0}{\ln(6/2)} \text{ F/m}$$

(iii) For imperfect dielectric, $R C = \frac{\epsilon}{\sigma} = \frac{3\epsilon_0}{5.13 \times 10^{-9}}$

$$R = \frac{1}{C} \times \frac{\epsilon}{\sigma}$$

$$= \frac{1}{C} \times \frac{3\epsilon_0}{5.13 \times 10^{-9}} \Omega/\text{m}$$

(iv) A leakage current can flow through this resistor.

$I_{\text{Leakage}}^2 R_{\text{Leakage}}$ is a power loss, which additional to the power loss of the cable resistance. The leakage current can cause other issues including of safety.

Question 4

(a) The relative permeability μ_r is a proportionality constant between magnetization vector $\vec{M}$ and flux density $\vec{B}$.

It is derived from the relation between magnetization vector $\vec{M}$, and field vectors $\vec{B}$ & $\vec{H}$

$$\vec{B} = \mu_0 \vec{H} + \mu_0 \vec{M}, \text{ but } \vec{M} \propto \vec{H}$$

$\vec{M} = \chi_m \vec{H}$, where χ_m : proportionality constant known as magnetic susceptibility.

Substituting,

$$\vec{B} = \mu_0 (1 + \chi_m) \vec{H} = \mu_0 \mu_r \vec{H}$$

$$\mu_r = 1 + \chi_m$$

Magnetization vector $\vec{M}$ is difficult to measure, as it relates to magnetic dipole moments at microscopic level. The relative permeability allows to measure ~~magnetization~~ the effect of magnetization in a material as is relative to free space (or vacuum),

(b) $\oint \vec{B} \cdot d\vec{l} = \mu NI$ $\mu = 5000 \mu_0$

$$B(2\pi r) = \mu NI$$

$$B = \frac{\mu NI}{2\pi r} = \frac{25}{1} T$$

$$\Phi = B \cdot S = 25 \times 7 \times 10^{-4} \text{ Wb}$$

(c) yes, airgap will affect the flux density, and flux density.

$$H_c l_c + H_g l_g = NI$$

$$\frac{B_c}{\mu} l_c + \frac{B_g}{\mu_0} l_g = NI$$

Some flux can leak away without crossing the airgap. Due to this leakage flux, the flux in the core and the gap may not be the same.

An air gap in the core is sometimes desirable as it increases the reluctance of the ~~air~~ magnetic ~~device~~ core, allowing more current in the circuit without magnetic saturation.

$$\Phi = \frac{NI}{R_c + R_g} = B \cdot S \quad \Bigg| \quad \Phi = \frac{NI}{R_c} = B \cdot S$$